

Steghide – Detailed Usage & Command Guide

1. What Steghide Does

Steghide hides (embeds) data inside files like images (JPG, BMP) or audio (WAV, AU). It can also encrypt and compress the hidden data.

2. Embed (Hide Data)

Basic usage:

```
steghide embed -cf cover.jpg -ef secret.txt
```

With output file:

```
steghide embed -cf cover.jpg -ef secret.txt -sf output.jpg
```

With password:

```
steghide embed -cf cover.jpg -ef secret.txt -p mypassword
```

Full example:

```
steghide embed -cf cover.jpg -ef secret.txt -sf steg.jpg -p pass123 -z 9
```

Flags:

-cf → cover file (carrier file)

-ef → file to embed

-sf → output stego file

-p → passphrase

-z → compression level (0–9)

3. Extract (Recover Data)

Basic extraction:

```
steghide extract -sf steg.jpg
```

Specify output file:

```
steghide extract -sf steg.jpg -xf output.txt
```

With password:

```
steghide extract -sf steg.jpg -p pass123
```

Flags:

-sf → stego file

-xf → extracted output file

-p → passphrase

4. Analyze File

Check if file has hidden data:

```
steghide info file.jpg
```

With password (if protected):

```
steghide info -p pass123 file.jpg
```

5. Encryption Options

List supported algorithms:

```
steghide encinfo
```

Common encryption algorithms:

aes128, aes192, aes256, rijndael-128, rijndael-256

6. Advanced Options

Disable compression:

```
steghide embed -cf cover.jpg -ef secret.txt -Z
```

Force overwrite:

```
steghide embed -cf cover.jpg -ef secret.txt -f
```

Verbose mode:

```
steghide embed -cf cover.jpg -ef secret.txt -v
```

Quiet mode:

```
steghide embed -cf cover.jpg -ef secret.txt -q
```

7. CTF Workflow

Step 1: Check file

```
steghide info suspicious.jpg
```

Step 2: Try extract (no password)

```
steghide extract -sf suspicious.jpg
```

Step 3: Try common passwords

```
steghide extract -sf suspicious.jpg -p rockyou
```

Step 4: Use brute force tools (e.g., stegseek)

```
stegseek suspicious.jpg /usr/share/wordlists/rockyou.txt
```